

Quantitative Data Analysis: instruction

A short exercise on interpreting quantitative data is required to be completed. This completed document should be uploaded as a standalone document with a naming convention as follows *name/supervisor/institution/competition/date*. The instruction is as follows.

Applicants applying for the SGSSS studentship entitled 'Understanding the complex place-based factors associated with the spread of COVID-19 infection amongst populations in the UK' are required to read this document and respond. The document describes a hypothetical example of a research dataset and three 'research questions' which might be addressed using these data. Applicants should provide responses to the three 'research questions' at the end of this document, using the response format specified. (NB the example of a research project outlined below is not directly related to the specific focus of the research which is the subject of the SGSSS project, but the example is intended to assess applicants' knowledge and skills in carrying out the kind of research which the SGSSS project will require.)

Background:

Birth weight is critically influential in early life and impacts long-term health and developmental outcomes. Public health and social science researchers continually examine factors contributing to variations in birth weight, such as maternal behaviours, demographic profiles, and health conditions. This exercise aims to explore the associations between various risk factors like ethnic background, maternal smoking during pregnancy, and birth weight.

Data Description:

The dataset, "birthweight.csv", consists of 169 entries, each with details spanning across ten variables related to birth conditions and maternal health.

Variables Included:

Low: Indicator (binary) if birth weight is less than 2.5 kg.
Age: Age of the mother in years.
Lwt: Mother's weight in pounds at the last menstrual period.
Race: Mother's race (1=white, 2=black, 3=other).
Smoke: Smoking status during pregnancy (binary).
Ptl: Number of previous premature labours.
Ht: History of hypertension (binary).
Ui: Presence of uterine irritability (binary).
Ftv: Number of physician visits during the first trimester.
Bwt: Birth weight in grams.

Research Questions:

Q1: Is there a difference in birth weights between mothers who smoked during pregnancy compared to those who did not?

- **Analytical method:** First, apply the Shapiro-Wilk test and Q-Q plot to assess the normality of the data of each group. If both are normally distributed, use an independent two-sample t-test to compare the mean birth weights of babies born to smoking and non-smoking mothers; otherwise, use the Mann-Whitney U test. Finally, calculate Cohen's d to assess the effect size. (57 words)

- **Rationale:** Despite the large sample size (169 entries), there may be a chance that one of the group is rather small or their distribution is especially skewed, which warrants a normality test. Parametric tests, such as the t-test, depend on the assumption of normality, or else one need to use the non-parametric equivalent. A minimal effect size, despite statistical significance, may have limited practical implications, so I recommend calculating Cohen's d to tell how different the difference is.
- **Interpretation:** As mentioned above, the chosen statistics test will indicate whether the difference is statistically significant, while calculated Cohen's d indicate how big is the difference.

Q2: Are there variations in birth weights among different racial groups?

- **Analytical method:** First, convert the numbers to categorical variables, which can be achieved in R with the 'factor' function. Next, assess the normality of birth weight distributions in each group using the Shapiro-Wilk test and the Q-Q plot. If all of them are normally distributed, perform a one-way ANOVA followed by post-hoc Tukey tests to identify specific group differences, or else opt for the non-parametric Kruskal–Wallis test with alternative post-hoc tests. (70 words)
- **Rationale:** Similarly, despite the large sample size, there may be a chance that one of the group is rather small or their distribution is especially skewed, which warrants a normality test. Parametric tests, such as one-way ANOVA, depend on the assumption of normality, or else one need to use the non-parametric equivalent. ANOVA or the Kruskal–Wallis test only indicate if differences exist, not which groups differ, which requires post-hoc analysis.
- **Interpretation:** As mentioned above, the chosen statistics test will indicate whether the difference is statistically significant, while post-hoc tests indicate which groups are different, providing clearer answers to the question. (It is also advisable to choose two groups from them to indicate how big the different, if necessary, as in Q1.)

Q3: Which demographic, health, and behavioural factors are most strongly associated with birth weight? Limit the analysis to a maximum of four variables.

- **Analytical method:** First, check the distribution and outliers of data, and convert skewed data through appropriate transformation (e.g. log transformation) and carefully determine whether to include the outliers or not. Second, create a set of reasonable choices of four variables. Third, implement multivariate regression on different choices of four variables and compare them by Akaike information criterion (AIC), with the lowest AIC indicating the optimal model. Four, interpret the optimal model to answer the question, while the other models may serve as sensitivity tests. (83 words)
- **Rationale:** Skewness may lead to concerns for non-normal residual distribution, which violates the assumption of ordinary least squares (OLS), leading to biased estimation. I often choose to either convert all data to log or not convert any variable for simplicity. The variable choices must be both practically reasonable while achieving best model performance. To avoid arbitrary selection, I tend to pre-select them and then compare their model AIC, which is a measure of information entropy, where lower values indicate better model choices in term of information losses. To ensure robustness, we can use unselected model as sensitivity tests to tell whether the estimation is robust across similar models.
- **Interpretation:** In the optimal model, factors with the largest significant coefficients are the most strongly associated with birth weight. One can also check whether the coefficients are close to each other in similar models to assess the robustness of estimation.

Reporting Format:

In response to each of these questions the applicant should submit a brief (50-100 words per question) description of the analytical method they would use to address the question, the rationale behind it, and a concise interpretation of how the results they would expect to produce would address the question.

I am not sure about whether the 100-word limit is applied to description of the analytical method only or for all of the answer. So I am here to provide a shorter version.

Q1: Is there a difference in birth weights between mothers who smoked during pregnancy compared to those who did not?

First, apply the Shapiro-Wilk test or Q-Q plots to determine the normality of the data. If normally distributed, use an independent two-sample t-test to compare the mean birth weights of babies born to smoking and non-smoking mothers; otherwise, use the Mann-Whitney U test. Optionally, calculate Cohen's d to assess the effect size. Parametric tests, such as the t-test, depend on the assumption of normality, or else one need to use the non-parametric Mann-Whitney U test. A minimal effect size, despite statistical significance, may have limited practical implications, so I recommend calculating Cohen's d to tell how different the difference is. (99 words)

Q2: Are there variations in birth weights among different racial groups?

First, convert the numbers to categorical variables. Next, assess the normality of birth weight distributions using the Shapiro-Wilk test. If normally distributed, perform a one-way ANOVA followed by post-hoc Tukey tests to identify specific group differences, or else opt for the nonparametric Kruskal-Wallis test with alternative post-hoc tests. Testing for normality ensures the correct method (parametric/nonparametric) is chosen. ANOVA or the Kruskal-Wallis test only indicate if differences exist, not which groups differ, which requires post-hoc analysis. Thus, one can tell which group differs, providing clearer answers to the question. (91 words)

Q3: Which demographic, health, and behavioural factors are most strongly associated with birth weight? Limit the analysis to a maximum of four variables.

First, consider the normality of data, and convert skewed data through appropriate transformation (e.g. log transformation). Second, implement multivariate regression on four variables and the selection of four variables can be compared by AIC/BIC, where lower values indicate better model choices. The optimal model should have a practically reasonable combination of variables plus low AIC/BIC. This approach assesses each factor's independent impact while controlling for others, highlighting the strongest predictors. In the optimal model, factors with the largest significant coefficients are the most strongly associated with birth weight. (88 words)